SEMESTER 2 • ENGINEERING SCIENCE • 45 PERIODS

Elementary Concepts of Electrical Systems

A complete handbook for Kerala Polytechnic Course 2032. This lesson expands the official syllabus into a textbook-style learning guide covering electrical safety, energy conservation, lamps, electric heating, electrolysis, batteries, engineering materials, electromagnetism, electrostatics and capacitor problems.

COURSE CODE
2032

PROGRAMME
Diploma in Electrical and Electronics Engineering

SEMESTER
2

CREDITS
3

CATEGORY
Engineering Science

PERIODS / WEEK
3 (L:3 T:0 P:0)

PERIODS / SEMESTER
45

PREREQUISITE
Applied Physics I + Mathematics I

Handbook method: each topic is explained as concept → diagram → application → formula → solved example → exam answer pattern.

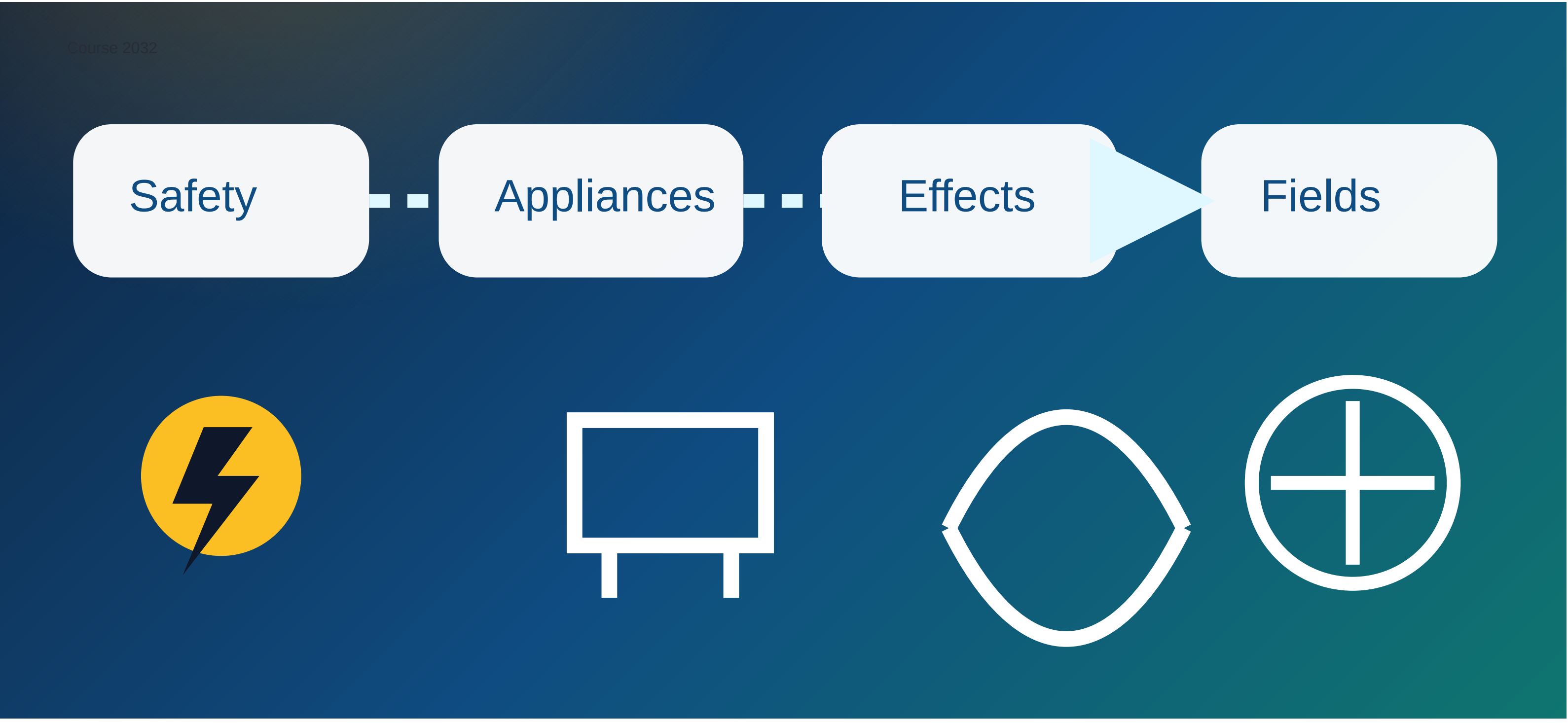
പഠനരീതി: ഈ subject electrical systems-ന്റെ basic idea ആണ്. Safety, appliances, lamps, heating, battery, magnetic field, static electricity, capacitor എന്നിവ concept-ആയി പഠിച്ചാൽ numerical problems എളുപ്പമാകും.

2032

From safety to fields

This handbook connects real home wiring and appliances with the physical effects of electric current and the field concepts used in electrical engineering.

Shock Safety | BEE Star Rating | Lamps | Heating | Batteries | Magnetism | Capacitors



OFFICIAL PDF CONTENT INCLUDED

Course objectives, prerequisites, outcomes and reference list

This section is added so the lesson is a complete handbook based on the supplied syllabus PDF, not only a summary of modules.

Course objectives

- ✓ Familiarize electrical safety measures and standards.
- ✓ Recognize lighting, heating and chemical effects of electric current.
- ✓ Provide knowledge of static and magnetic effects of electric current.
- ✓ Comprehend the concepts of electromagnetic effect.

Course prerequisites

Topic	Course name	Semester
Basic knowledge in Physics	Applied Physics I	1
Basic knowledge in Mathematics	Mathematics I	1

Course outcomes

CO	Description	Duration	Cognitive level
CO1	Choose various residential electrical appliances for energy conservation.	10 h	Applying
CO2	Summarize the luminous, heating and chemical effects of electric current.	11 h	Understanding
CO3	Classify the storage batteries and electrical engineering materials.	11 h	Understanding
CO4	Identify the basic concepts of electromagnetism and electrostatics.	11 h	Applying
Series tests	Two internal series tests.	2 h	—

Text / reference books and online resources from syllabus

Type	Book / resource
T1	B. L. Theraja, Electrical Technology Vol-I, S. Chand Publications.
T2	B. L. Theraja and A. K. Theraja, A Text Book of Electrical Technology Vol-III, S. Chand and Co.
R1	J. B. Gupta, A Course in Electrical Technology-I, S. K. Kataria & Sons.
R2	Guide Books No. 1 and 3 for National Certification Examination for Energy Managers and Energy Auditors, Bureau of Energy Efficiency.
R3–R10	V. K. Mehta and Rohit Mehta; Ashfaq Husain; Mittle and Mittal; Saxena S. B. Lal; Ritu Sahdev; V. Jegathesan; R. S. Sedha; CEA Safety Regulations.
Online	Electrical4U, NPTEL, Electrical Engineering Portal, All About Circuits, NPTEL course 108105112 and MIT OCW electrical engineering resources.

SYLLABUS STRUCTURE

Official course map converted into handbook sections

The official syllabus has 4 course outcomes and 43 teaching hours, plus 2 hours for series tests. This handbook keeps the same order and expands every module into readable explanations, examples and exam practice.

CO1 • 10 hours

Choose residential electrical appliances for energy conservation.

CO2 • 11 hours

Summarize luminous, heating and chemical effects of electric current.

CO3 • 11 hours

Classify storage batteries and electrical engineering materials.
Course 2032

CO4 • 11 hours

Identify basic concepts of electromagnetism and electrostatics.

Module	Outcome / syllabus focus	Sub-topics	Hours	Level
M1.01	Electric shock effects and precautions	Hazards, shock phenomenon, accident response	2	Understanding
M1.02	Safety devices and residential precautions	Fuse, ELCB, MCCB, RCCB, safe installation practice	3	Understanding
M1.03	Domestic appliance specifications and energy calculation	Lamp, fan, iron, mixer, heater, induction cooker, refrigerator, AC, pump, washing machine, TV, microwave, computer, UPS, inverter	3	Applying
M1.04	BEE star rating and energy conservation	Domestic tips, appliance selection, star rating criteria	2	Understanding
M2.01	Lamps	Incandescent, fluorescent, sodium vapour, LED, CFL, induction lamp	4	Understanding
M2.02–M2.03	Electric heating	Heating principle, heat transfer, resistance, induction, dielectric heating	5	Understanding
M2.04	Electrolysis	Principle, Faraday's laws, applications	2	Understanding
M3.01–M3.03	Storage batteries	Ratings, efficiency, lead-acid battery, Li-ion, series/parallel connection, charging	8	Understanding
M3.04	Electrical and magnetic materials	Conducting, insulating, ferromagnetic, diamagnetic, paramagnetic materials	3	Understanding
M4.01–M4.04	Magnetism, electromagnetism, electrostatics, capacitors	Laws, flux, permeability, Faraday, Lenz, inductance, field, potential, capacitance	11	Understanding / Applying

Exam preparation order: Safety definitions → appliance energy problems → lamp comparison → heating methods → battery ratings → material classifications → field formulas → capacitor numericals.

FORMULA AND KEY-POINT BANK

Minimum equations and facts to remember

Power and energy

$$P = VI = I^2R = V^2/R$$

$$\text{Energy in kWh} = (\text{Watt} \times \text{Hours}) / 1000$$

$$\text{Bill amount} = \text{Units} \times \text{Rate per unit}$$

Use appliance rating plate values for domestic energy calculation.

Heating effect

$$H = I^2Rt \text{ joule}$$

$$1 \text{ calorie} \approx 4.186 \text{ joule}$$

Heating increases with square of current, resistance and time.

Electrolysis

$$m = ZIt$$

$$m \propto Q, \text{ where } Q = It$$

Mass deposited depends on current and time.

Battery

$$\text{Ah} = \text{Current} \times \text{Time}$$

$$\text{Wh} = \text{Voltage} \times \text{Ah}$$

$$\eta\text{Ah} = \text{Output Ah} / \text{Input Ah} \times 100\%$$

$$\eta\text{Wh} = \text{Output Wh} / \text{Input Wh} \times 100\%$$

Magnetism and inductance

$$B = \Phi / A$$

$\mu r = \mu / \mu 0$

$e = -N\, d\Phi/dt$

Energy in inductor = $\frac{1}{2}LI^2$

Electrostatics and capacitors

$C = Q / V$

Energy in capacitor = $\frac{1}{2}CV^2$

Parallel: $Ceq = C1 + C2 + C3 + \ldots$

Series: $1/Ceq = 1/C1 + 1/C2 + 1/C3 + \ldots$

Common mistake: kW and W are not the same. Always convert watt to kilowatt before calculating kWh.

MODULE 1 • CO1 • 10 HOURS

Electrical safety, domestic appliances and energy conservation

Outcome: choose residential electrical appliances for energy conservation.

M1.01 Electric shock

Electric shock occurs when current passes through the human body. Severity depends on current magnitude, path through body, contact time, body resistance, voltage, frequency and condition of skin.

- ✓ Dry skin gives higher resistance; wet skin gives lower resistance.
- ✓ Current through chest is more dangerous because it affects heart and lungs.
- ✓ Never touch a victim before isolating the supply.

> Shock കിട്ടുമ്പോൾ ആദ്യം supply OFF ചെയ്യണം. Victim-നെ direct ആയി തൊടരുത്.

M1.02 Safety devices

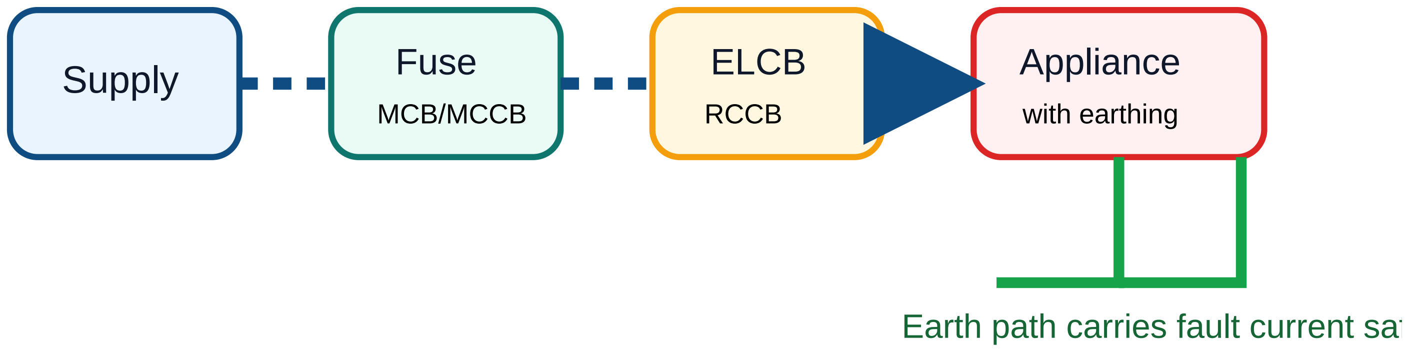
Safety devices interrupt abnormal current or earth leakage before damage or injury becomes severe.

Device	Purpose	Key idea
Fuse	Over-current protection	Fuse element melts when current exceeds rated value.
MCB / MCCB	Automatic circuit breaking	Trips during overload or short circuit.
ELCB / RCCB	Earth leakage protection	Trips when leakage current flows to earth.
Earthing	Shock protection	Provides low-resistance path for fault current.

M1.03 Appliance specifications

Domestic appliances are identified by voltage, wattage, current, frequency, capacity, duty, energy rating and safety markings.

Examples: lamp, ceiling fan, iron box, mixer grinder, heater, induction cooker, refrigerator, air conditioner, domestic pump, washing machine, TV, microwave oven, computer, UPS and inverter.



Energy conservation in domestic buildings

Lighting

Use LED lamps, daylighting, proper switching and task lighting. LED has high efficacy and low heat loss compared to incandescent lamps.

Refrigerator

Choose correct capacity, avoid frequent door opening, keep condenser clean and allow ventilation around the unit.

Air conditioner

Select proper tonnage, use BEE star rating, close air leaks and set thermostat reasonably.

Course 2032

Standby loads

Switch off chargers, TVs, computers and UPS loads when not needed.

BEE star rating

BEE star rating helps consumers select energy-efficient appliances. More stars generally indicate lower energy consumption for comparable capacity and usage category.

Answer pattern: Define BEE → explain star label → mention energy consumption comparison → give domestic example → conclude with conservation benefit.

Solved examples

Problem 1: A 750 W iron box is used for 2 hours per day for 20 days. Find energy consumed.

Solution: Power = 750 W = 0.75 kW. Time = 2 × 20 = 40 h. Energy = 0.75 × 40 = **30 kWh**.

Problem 2: A 1.5 kW water heater is used for 2 hours per day for 30 days. Find energy consumed and cost at ₹6/unit.

Solution: Energy = 1.5 × 2 × 30 = **90 kWh**. Cost = 90 × 6 = **₹540**.

Expected questions — Module 1

Q1. Explain electric shock and list precautions against electric shock.

Q2. Compare fuse, MCB, MCCB, ELCB and RCCB.

Q3. A 1.5 kW heater is used for 3 hours daily for 15 days. Calculate energy consumption.

Q4. Explain BEE star rating and its importance in domestic appliance selection.

Q5. Write energy conservation tips for lighting, refrigerator and television.

MODULE 2 • CO2 • 11 HOURS

Luminous, heating and chemical effects of electric current

Outcome: summarize the luminous, heating and chemical effects of electric current.

Electrical lamps

Incandescent lamp

Light is produced when a tungsten filament becomes white hot due to current. It is simple but inefficient because much energy

is lost as heat.
Course 2032

Fluorescent and CFL

Mercury vapour discharge produces ultraviolet radiation, which is converted to visible light by phosphor coating. CFL is a compact form.

Sodium vapour lamp

Produces yellowish light and is used for street lighting. Low-pressure and high-pressure sodium lamps differ in construction and colour rendering.

LED lamp

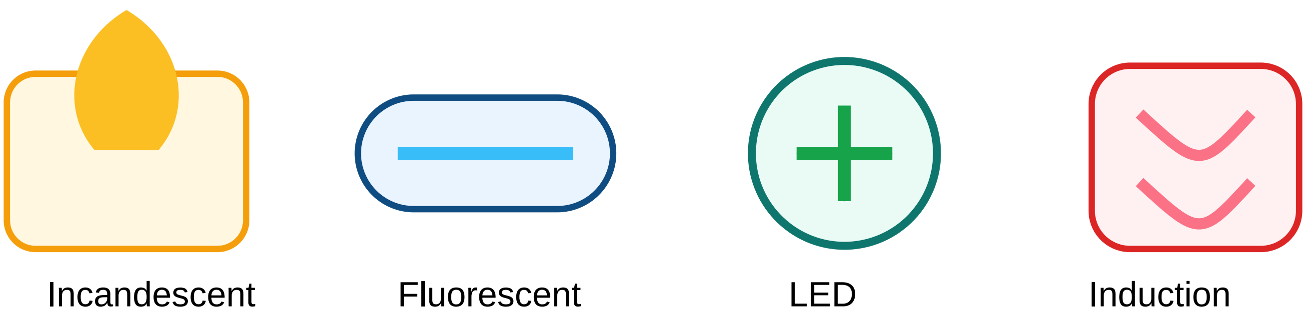
Light Emitting Diode converts electrical energy directly into light by electroluminescence. It is efficient, long life and suitable for domestic lighting.

Induction lamp

Uses electromagnetic induction to excite gas discharge without conventional electrodes. It has long life and is used in special lighting.

Selection idea

For domestic energy conservation, LED lamps are generally preferred because of high efficacy and lower wattage for same illumination.



Electric heating

Electric heating is based mainly on Joule's law of heating. When current passes through resistance, electrical energy is converted into heat.

$$H = I^2Rt$$

Modes of heat transfer

- ✓ Conduction: heat transfer through solid contact.
- ✓ Convection: heat transfer by fluid motion.
- ✓ Radiation: heat transfer by electromagnetic waves.

Good heating element

- ✓ High resistivity
- ✓ High melting point
- ✓ Low temperature coefficient
- ✓ Oxidation resistance
- ✓ Mechanical strength

Advantages

- ✓ Clean operation
- ✓ Easy temperature control
- ✓ High efficiency at point of use
- ✓ Uniform heating possible
- ✓ No flue gas

Methods of electric heating

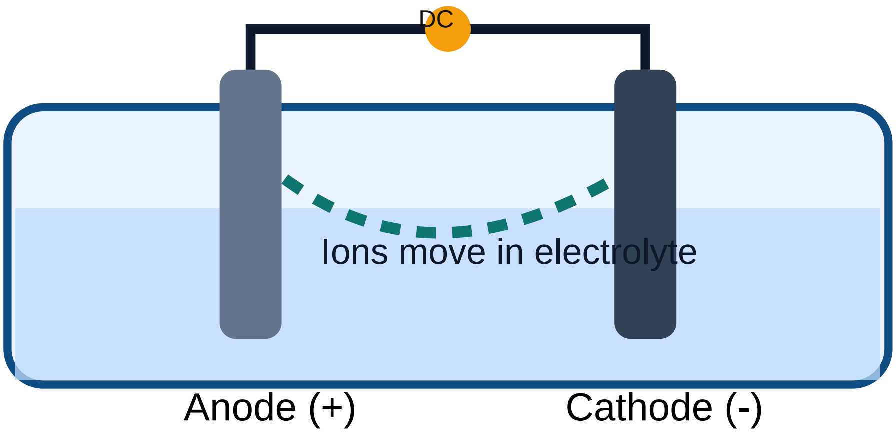
Method	Principle	Example / use
Direct resistance heating	Current passes through charge itself.	Salt bath, electrode boiler principle.
Indirect resistance heating	Current heats element; heat transfers to charge.	Room heater, oven, iron box.
Induction heating	Eddy currents are induced in conducting material.	Induction furnace, induction cooker.
Dielectric heating	Dielectric loss heats insulating material in high-frequency field.	Plastic, wood, drying applications.

Electrolysis

Electrolysis is the chemical decomposition of an electrolyte by passing direct current through it. It is used in electroplating, refining, extraction and battery processes.

$$\text{Faraday's first law: } m \propto Q = It$$

$$m = ZIt$$



Solved example: heating

Problem: A 5 Ω heater carries 8 A for 10 minutes. Find heat produced.

Solution: $H = I^2Rt = 8^2 \times 5 \times 600 = 64 \times 3000 =$ **192000 J**.

Expected questions — Module 2

Q1. Explain construction and working of LED lamp.

Q2. Compare incandescent, fluorescent, CFL, sodium vapour and LED lamps.

Q3. State the requirements of a good heating element.

Q4. Explain direct resistance, indirect resistance, induction and dielectric heating.

Q5. State Faraday's laws of electrolysis and mention applications.

MODULE 3 • CO3 • 11 HOURS

Storage batteries and electrical engineering materials

Outcome: classify storage batteries and electrical engineering materials.

Battery basics

Cell and battery

A cell converts chemical energy into electrical energy. A battery is a group of cells connected to obtain required voltage or

capacity.
Course 2032

Primary and secondary cells

Primary cells are not normally rechargeable. Secondary cells are rechargeable and used for storage batteries.

Ratings

Common battery ratings include voltage, Ah capacity, Wh energy, charge/discharge current, efficiency and life cycle.

$$\text{Battery capacity in Ah} = \text{Current} \times \text{Time}$$

$$\text{Energy in Wh} = \text{Voltage} \times \text{Ah}$$

Lead-acid battery

Main parts

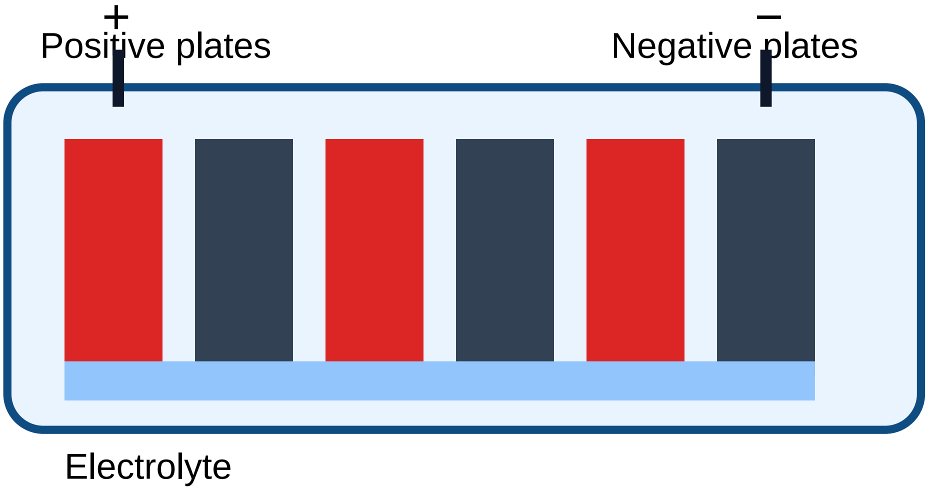
- ✓ Positive plate: lead dioxide
- ✓ Negative plate: spongy lead
- ✓ Electrolyte: dilute sulphuric acid
- ✓ Separators, container, vent plug, terminals

Fully charged indications

- ✓ Cell voltage reaches normal charged value.
- ✓ Specific gravity becomes steady.
- ✓ Gassing occurs near end of charge.
- ✓ Electrolyte condition becomes stable.

Care and maintenance

- ✓ Keep terminals clean and tight.
- ✓ Maintain proper electrolyte level.
- ✓ Avoid overcharging and deep discharging.
- ✓ Provide ventilation during charging.



Series and parallel connection of batteries

Series connection

Positive terminal of one battery is connected to negative terminal of the next. Voltage adds; Ah capacity remains same.

$$V_{total} = V1 + V2 + V3 + ...$$

Parallel connection

All positive terminals are connected together and all negative terminals together. Voltage remains same; Ah capacity adds.

$$Ah_{total} = Ah1 + Ah2 + Ah3 + ...$$

Lithium-ion battery

Lithium-ion batteries are rechargeable, high energy density and widely used in mobile phones, laptops, electric vehicles, UPS systems and portable equipment. They need protection circuits to avoid overcharge, over-discharge and overheating.

Electrical engineering materials

Class	Properties	Examples	Applications
Conducting materials	Low resistivity, high conductivity	Copper, aluminium	Wires, busbars, windings
Insulating materials	High resistivity, high dielectric strength	PVC, rubber, porcelain, mica, glass	Cable insulation, switchgear, supports
Ferromagnetic	Strongly attracted, high permeability	Iron, nickel, cobalt	Transformer core, machines
Paramagnetic	Weakly attracted	Aluminium, platinum	Special applications
Diamagnetic	Weakly repelled	Copper, bismuth, water	Material classification

ഓർമ്മിക്കാൻ: Conductor current carry ചെയ്യും, insulator current തടയും, ferromagnetic material magnetic field ശക്തമാക്കും.

Solved example: battery capacity

Problem: A 12 V, 40 Ah battery supplies 4 A load. Estimate ideal operating time.

Solution: Time = Ah / A = 40 / 4 = **10 hours**. Practical time is lower due to losses and discharge limits.

Expected questions — Module 3

Q1. Differentiate primary and secondary batteries.

Q2. Explain ampere-hour efficiency and watt-hour efficiency.

Q3. Draw and label the parts of a lead-acid battery.

Q4. Compare series and parallel battery connections.

Q5. Classify electrical materials and magnetic materials with examples.

MODULE 4 • CO4 • 11 HOURS

Magnetism, electromagnetism, electrostatics and capacitors

Outcome: identify basic concepts of electromagnetism and electrostatics and solve simple capacitor problems.

Magnetism

Laws of magnetism

Like poles repel and unlike poles attract. Magnetic lines of force emerge from north pole and enter south pole outside the magnet.

Important terms

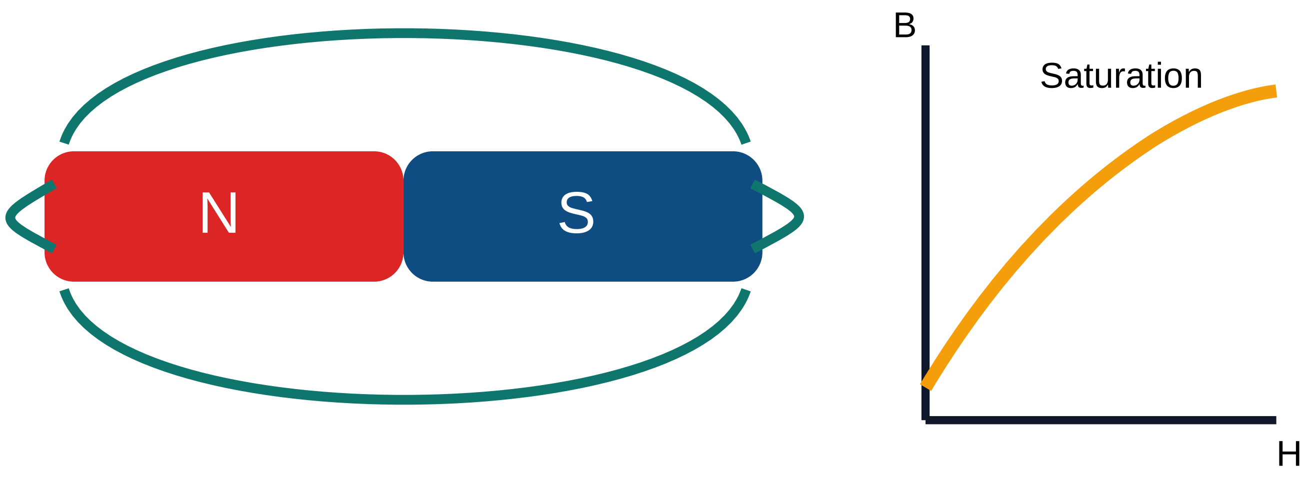
- ✓ Magnetic flux Φ: total magnetic lines.
- ✓ Flux density B: flux per unit area.
- ✓ Field strength H: magnetising force.
- ✓ Permeability μ: ability to allow magnetic flux.

B-H curve

The B-H curve shows relation between flux density and magnetising force. It indicates saturation, retentivity and hysteresis behaviour of magnetic material.

$$B = \Phi / A$$

$$\mu r = \mu / \mu 0$$



Electromagnetism

When current flows through a conductor, magnetic field is produced around it. When magnetic flux linked with a coil changes, emf is induced in the coil.

Faraday's law: induced emf is proportional to rate of change of flux linkage.

$$e = -N\, d\Phi/dt$$

Lenz's law: The direction of induced emf is such that it opposes the cause producing it.

Self inductance

Property by which a coil opposes change of current in itself due to induced emf.

Mutual inductance

Property by which change of current in one coil induces emf in another nearby coil.

Stored energy

$$W = \tfrac{1}{2}LI^2$$

Electrostatics

Electrostatics deals with electric charges at rest, electric field, electric flux, potential and permittivity.

Term	Meaning
Electric flux	Total electric lines of force from a charged body.
Electric field	Region where a charge experiences force.
Field intensity	Force experienced by unit positive charge.
Electric potential	Work done per unit charge to bring a charge from infinity.
Permittivity	Ability of medium to permit electric flux.

Capacitors

A capacitor stores electric charge and energy in an electric field. It consists of two conducting plates separated by dielectric material.

$$C = Q / V$$

$$\text{Energy stored} = \tfrac{1}{2}CV^2$$

Series capacitors

Same charge flows; voltage divides. Equivalent capacitance is less than the smallest capacitance.

$$1/C_{eq} = 1/C_1 + 1/C_2 + \ldots$$

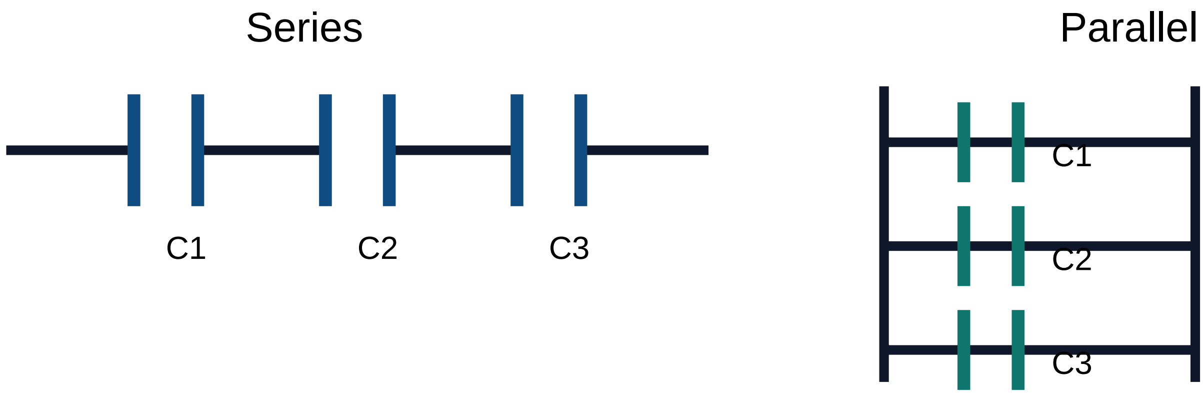
Parallel capacitors

Same voltage appears across each capacitor. Equivalent capacitance is the sum of individual capacitances.

$$C_{eq} = C_1 + C_2 + \ldots$$

Applications

Filtering, coupling, power factor improvement, timing circuits, motor starting and energy storage.



Solved example: capacitor equivalent

Problem: Find equivalent capacitance of 6 μF and 3 μF in series.

Solution: $1/C = 1/6 + 1/3 = 1/6 + 2/6 = 3/6 = 1/2$. Therefore $C =$ **2 μF**.

Problem: Find equivalent capacitance of 6 μF and 3 μF in parallel.

Solution: $C = 6 + 3 =$ **9 μF**.

Expected questions — Module 4

Q1. State laws of magnetism and define flux, flux density and permeability.

Q2. Explain B-H curve and its importance.

Q3. State Faraday's laws and Lenz's law.

Q4. Compare electric circuit and magnetic circuit.

Q5. Derive equivalent capacitance for capacitors in series and parallel.

QUICK REVISION

One-page memory sheet

Safety

Shock depends on current, path and time. Isolate supply before rescue. Fuse/MCB protects from over-current. RCCB/ELCB protects from leakage.

Course 2032

Energy

Unit = kWh. Energy = kW × hours. More BEE stars means more energy efficiency for the same appliance category.

Lamps and heating

LED is efficient. Heating follows $H = I^2Rt$. Induction heating uses eddy currents. Electrolysis follows $m = ZIt$.

Batteries

Series increases voltage. Parallel increases Ah capacity. Lead-acid battery uses PbO_2, Pb and dilute H_2SO_4.

Materials

Conductors carry current. Insulators oppose current. Ferromagnetic materials are strongly attracted and have high permeability.

Fields

$B = \Phi/A$, $e = -N\, d\Phi/dt$, $W_L = \frac{1}{2}LI^2$, $C = Q/V$, $W_C = \frac{1}{2}CV^2$.

MODEL QUESTION PAPER

Official-pattern practice paper with complete answer key below

This is an original practice paper based on the syllabus order. It does not copy official questions.

COURSE 2032 — ELEMENTARY CONCEPTS OF ELECTRICAL SYSTEMS
Time: 2 Hours   Maximum Marks: 50

PART A — Very short answer questions. Answer all. (10 × 2 = 20)
1. Define electric shock.
2. Mention two precautions against electric shock.
3. What is the function of RCCB?
4. Define one unit of electrical energy.
5. State one advantage of LED lamp.
6. State Joule's law of heating.
7. Define ampere-hour rating of a battery.
8. Name two conducting materials.
9. State Faraday's law of electromagnetic induction.
10. Write the formula for equivalent capacitance of capacitors in parallel.

PART B — Short answer questions. Answer any five. (5 × 4 = 20)

11. Explain procedure to be followed when an electric shock accident occurs.
12. Compare fuse, MCB and RCCB.
13. Explain BEE star rating of electrical appliances.
14. Compare incandescent lamp and LED lamp.
15. Explain direct and indirect resistance heating.
16. Explain constructional parts of a lead-acid battery.
17. Classify magnetic materials with examples.
18. Derive equivalent capacitance of two capacitors connected in series.

PART C — Long answer / problem questions. Answer any two. (2 × 5 = 10)

19. A 1.5 kW water heater is used for 2 hours per day for 30 days. Calculate energy consumed and cost at ₹6 per unit.
20. Explain Faraday's laws of electrolysis and electromagnetic induction. Mention one application of each.
21. Explain laws of magnetism, B-H curve and importance of permeability.
22. Two capacitors of 4 μF and 6 μF are connected (a) in parallel and (b) in series. Find equivalent capacitance.

MASTER ANSWER KEY

Complete answers for expected questions and model paper

Module answers

Module 1 answers

Electric shock: Electric shock is the physiological effect produced when electric current passes through the body. Precautions include switching off supply before work, using insulated tools, dry hands, proper earthing, RCCB/ELCB protection, avoiding damaged cords and following lock-out procedure. During accident, isolate supply, separate victim with dry insulating material, call medical help and give first aid/CPR if trained.

Energy problem: For a 1.5 kW heater used 3 h/day for 15 days: Energy = 1.5 × 3 × 15 = 67.5 kWh.

BEE star rating: It is an energy-efficiency label given by Bureau of Energy Efficiency. It allows comparison of appliances of same category. Higher star rating indicates better energy efficiency and lower operating cost.

Module 2 answers

LED lamp: LED produces light by electroluminescence when a forward-biased p-n junction emits photons. It has high efficiency, long life, low heat, fast switching and compact size. It is preferred for domestic lighting.

Heating methods: Direct resistance heating passes current through charge itself. Indirect resistance heating uses a heating element. Induction heating produces eddy currents in conductor. Dielectric heating heats insulating materials by dielectric loss in high-frequency field.

Electrolysis: Electrolysis is chemical decomposition of electrolyte by DC. Faraday's first law states mass deposited is proportional to quantity of electricity. $m = ZIt$. Applications include electroplating, refining and extraction of metals.

Module 3 answers

Lead-acid battery: It has positive plates of lead dioxide, negative plates of spongy lead, dilute sulphuric acid electrolyte, separators, container, terminals and vent plugs. Care includes maintaining electrolyte level, cleaning terminals, avoiding overcharge/deep discharge and charging in ventilated place.

Series/parallel batteries: Series connection increases voltage while Ah remains same. Parallel connection keeps voltage same while Ah capacity adds.

Materials: Conductors such as copper and aluminium have low resistivity. Insulators such as PVC, porcelain, mica and rubber have high resistivity. Ferromagnetic materials are strongly attracted, paramagnetic weakly attracted and diamagnetic weakly repelled.

Module 4 answers

Faraday and Lenz laws: Faraday's law says emf is induced when flux linked with a circuit changes and magnitude is proportional to rate of change of flux linkage. Lenz's law gives direction: induced emf opposes the cause producing it.

Capacitors: For parallel capacitors, $C = C1 + C2 + \dots$. For series capacitors, $1/C = 1/C1 + 1/C2 + \dots$. Energy stored in capacitor is $\frac{1}{2}CV^2$.

B-H curve: It shows relationship between flux density B and magnetising force H. It helps understand permeability, saturation, retentivity, coercivity and suitability of magnetic materials.

Model paper answer key

1. Electric shock is the effect of current passing through the human body.

2. Use insulated tools and switch off/isolate supply before touching electrical parts.

3. RCCB trips the circuit when earth leakage/residual current is detected.

4. One unit = 1 kWh, energy used by 1 kW load for 1 hour.

5. LED lamp is energy efficient and has long life.

6. Heat produced in a conductor is $H = I^2Rt$.

7. Ah rating indicates current a battery can supply for a specified time: $Ah = I \times t$.

8. Copper and aluminium.

9. Induced emf is proportional to rate of change of flux linkage: $e = -N\, d\Phi/dt$.

10. $C_{eq} = C1 + C2 + C3 + \ldots$

11–18. Use the module answers above. Write definition, principle, diagram/table and application where suitable.

19. Energy = 1.5 × 2 × 30 = 90 kWh. Cost = 90 × 6 = ₹540.

20. Faraday's electrolysis law: $m = ZIt$; application electroplating. Electromagnetic induction: $e = -N\, d\Phi/dt$; application generator/transformer.

21. Like poles repel, unlike poles attract. B-H curve shows magnetisation characteristics and saturation. Permeability indicates how easily material carries magnetic flux.

22. Parallel: C = 4 + 6 = 10 μF. Series: 1/C = 1/4 + 1/6 = 5/12, so C = 12/5 = 2.4 μF.